



ROOT CAUSE UNLOCKED

Unlock Your Brain *Health*

Clearer Thinking, a Steadier Mood, and the Treatable Causes Worth Ruling Out First

A guide for adults noticing brain fog, memory that feels less sharp than it used to be, or persistent anxiety and low mood, who want to understand the terrain underneath those symptoms, alongside whatever medical or mental health care they are already receiving.

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BEFORE YOU START

A note before you start. This guide is educational information about the biology behind brain fog and mood symptoms, and the general categories of evaluation and support used to address treatable contributors to both. It is not individualized medical, psychiatric, or psychological advice, it does not diagnose any condition, and it does not replace your relationship with your physician, psychiatrist, therapist, or my team. If you are currently in therapy or under psychiatric care, please continue exactly as prescribed and treat everything here as additive, not a replacement. Before anything else, here is what to do if you are in crisis right now.

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If You Are in Crisis Right Now

If you are having thoughts of harming yourself or of suicide, please stop reading and reach out immediately. Call or text **988** to reach the Suicide and Crisis Lifeline in the United States, available 24 hours a day, every day. If you or someone else is in immediate danger, call **911** or go to your nearest emergency department right now.

You do not need a diagnosis, a plan, or a “good enough” reason to reach out. You do not need to have read another word of this book. This applies exactly as much if your symptoms feel mild and confusing as it does if they feel severe.

If you are worried about a friend or family member, the same resources apply to you. You can call or text 988 to ask how to support someone else, or call 911 if there is immediate danger.

Everything else in this book can wait. This cannot.

Red Flags That Need Same-Day Medical Attention, Not a Supplement Plan

Most of the brain fog and mood symptoms this book addresses build gradually, over weeks or months, and are safe to explore with the terrain-based approach described here. A few patterns are different, and they deserve an emergency department, an urgent care visit, or a same-day call to a physician instead of a supplement plan or a symptom tracker. Please seek prompt medical evaluation if you or someone you love experiences:

- **Sudden or rapid confusion, memory loss, or personality change** over hours to a few days. This pattern can indicate a stroke, a brain bleed, an infection affecting the brain, medication toxicity, or a condition called normal pressure hydrocephalus, all of which need urgent evaluation, not a root-cause workup.
- **Any new focal neurologic sign:** weakness or numbness on one side of the body, slurred speech, sudden vision loss, loss of coordination, or a severe headache unlike any you have had before.
- **Psychosis** (hearing or seeing things that are not there, or fixed false beliefs) **or mania** (a dramatically decreased need for sleep, racing thoughts, grandiosity, or impulsive behavior that is out of character).
- **Any thoughts of suicide or of harming yourself or someone else.** See the crisis section above. This is always urgent, regardless of severity or duration.
- **Confusion or cognitive change accompanied by fever, a recent head injury, or a fall.**
- **A sudden, marked change in cognition in someone taking multiple medications**, especially after starting a new prescription or changing a dose.

These situations are emergencies or urgent medical evaluations. A functional or root-cause approach is not the right first step for any of them, and none of them are something to wait out or address with the material that follows.

On Psychiatric Medication: Read This Before Anything Else

If you take an antidepressant, an anti-anxiety medication, a mood stabilizer, or any other psychiatric prescription, this book will never tell you to stop, reduce, skip a dose, or change how you take it. That decision belongs to you and the physician or psychiatric provider managing your care, full stop. Nothing in the pages that follow is designed to create an opening for stopping a prescription on your own, and if any sentence in this book ever reads that way, it does not reflect what I intend and should be disregarded in favor of your prescriber's guidance.

This is worth taking seriously, not as a formality. Stopping an SSRI or SNRI antidepressant abruptly, or missing doses irregularly, can cause a real and well-documented discontinuation syndrome: dizziness, electric-shock sensations in the head, flu-like symptoms, irritability, sleep disruption, and a return of the original depression or anxiety, sometimes more intensely than before. A 2025 meta-analysis pooling data across 35 studies found that withdrawal symptoms occur in roughly 43 percent of people who stop an antidepressant, and around 46 percent of people specifically stopping an SSRI, usually appearing within the first two weeks.¹ Reducing the dose gradually under a prescriber's supervision lowers that risk compared with stopping all at once, but it does not eliminate it.¹ Any adjustment to your prescription, including a gradual, medically supervised reduction, is a conversation to have with the person who prescribed it. It is never something to start on your own based on a book, a supplement label, a symptom quiz, or a feeling that you are ready.

None of this means medication is required forever, and it does not mean the approach in this book has nothing to offer you. It means that if a change to your prescription ever makes sense, your prescriber is the person to make that decision with. Everything that follows is written to run alongside psychiatric and medical care, not instead of it. If you are also seeing a therapist or psychiatrist, that relationship is not a placeholder until a root-cause approach "kicks in." For many people it is the most important part of the care they are receiving, and this book's job is to support it, not compete with it.

The Symptoms Nobody Quite Explains

You walk into a room and forget why. A word you have used a thousand times sits just out of reach. You read the same paragraph three times and still cannot tell anyone what it said. Meetings that used to be easy now take real effort to follow. And underneath the fog, or sometimes instead of it, there is a low mood or a low-grade anxiety that does not seem to have an obvious cause, that does not lift with a good night of sleep, and that makes the fog itself feel heavier.

If you have taken this to a doctor, you may have heard some version of “your labs are normal” or “this is probably just stress, aging, or a busy season of life.” Sometimes that is genuinely the full answer. Very often it is not the full answer, because standard bloodwork screens for a narrow set of conditions and stops there, and a fifteen-minute visit rarely has room for the kind of layered, connected questioning that brain fog and mood symptoms actually require.

This book is written for two overlapping groups of people. The first is anyone noticing brain fog: word-finding trouble, difficulty concentrating, forgetfulness that feels new, or a general sense that your thinking is not as sharp as it used to be, without a diagnosis of any neurodegenerative disease. The second is anyone living with anxiety or depression who wants to understand what is happening in the body underneath the mood, as a complement to whatever mental health care they are already receiving or considering.

This book is explicitly **not** written for, and does not address, diagnosed dementia, Alzheimer’s disease, or Parkinson’s disease. Those are conditions that deserve ongoing care from a neurologist and a dementia care team, not a root-cause reframing, and if that describes your situation or a loved one’s, please treat this book as background context at most, not a plan. If your primary concern is ADHD, stimulant medication and dosing decisions belong with your prescriber, and this book covers ADHD-adjacent brain fog only where it overlaps the terrain below. If migraines, vertigo, POTS, or lingering post-viral fatigue are your main complaint, our companion guides *Unlock Your Migraines* and *Unlock Your Post-Viral Recovery* address those territories directly. If persistent physical fatigue and low energy, rather than thinking clearly or your mood, are the dominant symptom, our companion guide *Unlock Your Energy* goes deeper into mitochondrial function and the HPA axis as energy drivers; this book only covers that overlap where it bears directly on cognition and mood.

Below are nine keys. Each one explains a piece of the terrain underneath brain fog and mood symptoms, the mechanism behind it in plain language, and what to consider doing about it at an educational level. The keys build toward a staged roadmap and two practical worksheets near the end, including a checklist to bring to your next medical appointment.

Brain Fog and Low Mood Are Signals, Not Verdicts

The problem nobody mentions. Brain fog and low mood are usually treated as endpoints: something you simply have, the way you have a certain height. In a root-cause framework, they are almost always downstream signals instead, produced by one or more upstream drivers working on the nervous system at the same time. Those drivers include gut function, inflammation, blood sugar regulation, sleep quality, nutrient status, chronic stress and HPA axis output, and mitochondrial energy production. When several of these are mildly off at once, none of them may look dramatic on a single lab result, but together they can produce exactly the fog and low mood you are living with.

The mechanism, in plain terms. Your brain is metabolically the most demanding organ in your body relative to its size, and it has almost no tolerance for interruptions in nutrient supply, blood sugar stability, inflammatory signaling, or sleep-dependent maintenance. Because so many systems feed into brain function, brain fog and mood are two of the least specific symptoms in medicine: they can be the first sign of dozens of underlying issues, most of which are common and treatable, and a smaller number of which need urgent attention (see the red flags above). That lack of specificity is exactly why brain fog and mood are so often dismissed. A single test rarely tells the whole story, so it is tempting to conclude nothing is wrong, or that what is wrong is simply “stress” or “getting older,” when in fact several small, real, correctable things are adding up.

What to consider, at an educational level. This book has no interest in convincing you the problem is “all in your head” versus “all in your body.” Its goal is to walk through the specific, identifiable upstream drivers that commonly produce brain fog and mood symptoms, starting with the ones that are the most straightforward to test and the most satisfying to fix, because finding a genuinely reversible contributor is the single most valuable thing a root-cause workup can hand you. Key 2 starts there.

The Treatable Causes Worth Ruling Out First

The gap in a standard workup. Every item on this list is common, testable with standard labs, and genuinely correctable, often without any specialized functional testing at all. A root-cause approach is at its strongest exactly here: pointing someone toward a proper workup that finds a reversible cause of their cognitive or mood symptoms is more valuable than any supplement protocol.

B12 and folate. Vitamin B12 deficiency produces a well-documented range of neuropsychiatric symptoms, including depression, cognitive slowing, memory difficulty, and in more severe or prolonged cases, more dramatic presentations.² It is common in older adults, in anyone with reduced stomach acid or malabsorption (including long-term use of acid-reducing medications), in strict vegans and vegetarians, and after certain bariatric surgeries. Low folate status has also been linked to depression risk. A 2026 analysis of national U.S. survey data in women found the relationship was not simply “more is better”: both the lowest and the highest quartiles of red blood cell folate were associated with higher odds of depression compared with a mid-range level, an important reminder that nutrient status is about sufficiency, not maximizing a number.³ Ask for a B12 level (ideally with methylmalonic acid, a more sensitive marker) and a folate level, ideally red blood cell folate rather than serum alone.

Thyroid function. Hypothyroidism, whether overt or subclinical, is a recognized cause of a genuine, distressing symptom complex that clinicians increasingly call “brain fog”: fatigue, low mood, and difficulty with memory and executive function that often predates the formal diagnosis.⁴ There is a complication worth knowing about. A large 2021 analysis that pooled individual data from more than 74,000 adults across 21 cohorts found that subclinical hypothyroidism and hyperthyroidism were not associated with measurable differences in cognitive test scores, cognitive decline, or dementia risk at the population level, and it did not support routinely screening for subclinical thyroid dysfunction specifically to explain cognitive decline.⁵ Both things can be true at once: brain fog is a real and recognized experience for many people with thyroid dysfunction, and large population studies have not consistently found it on standardized cognitive testing, particularly for milder, subclinical thyroid abnormalities. A TSH and free T4 are simple, low-cost tests either way, and correcting a genuinely abnormal thyroid level is safe and often helps, even where the population-level cognitive testing data is more mixed than commonly assumed.

Obstructive sleep apnea. Loud snoring, witnessed pauses in breathing, gasping awake, or persistent daytime sleepiness deserve a real conversation about a sleep study, not just a comment about needing more rest. A 2026 study of nearly 2,800 cognitively unimpaired middle-aged adults found that self-reported obstructive sleep apnea was associated with poorer memory and a higher composite dementia-risk score, regardless of genetic risk status.⁶ Sleep apnea is also directly treatable, which makes it one of the higher-value items on this list to rule out.

Anemia. Low hemoglobin reduces the oxygen available to your brain, and the relationship to cognition is measurable even at levels well above what most people consider dangerously low. A 2026 study of

hospitalized older adults found that each 1 g/dL drop in hemoglobin was associated with a meaningful worsening in a standard cognitive screening score, an effect that was two to three times larger in people over 65 than in younger adults.⁷ A basic complete blood count and ferritin level will catch most cases.

Vitamin D and other B vitamins. Vitamin D deficiency and low intake of B vitamins including thiamin, riboflavin, niacin, and folate have each been linked to depressive symptoms, though a 2024 review of the evidence was candid that study designs vary widely and vitamin D supplementation trials for mood specifically have produced inconsistent results, which puts the fair summary at “worth checking and correcting if low,” not “guaranteed to fix your mood.”⁸ A vitamin D (25-OH) level is inexpensive and worth including with any of the labs above.

Medications, including over-the-counter ones. Many common medications, especially those with anticholinergic activity (some allergy medications, some bladder medications, some older antidepressants, some sleep aids), are associated with measurable cognitive impairment, particularly as the total anticholinergic burden across several medications adds up. A 2026 study using a validated anticholinergic burden scale found that people with a high anticholinergic burden had meaningfully more cognitive impairment than those with a low burden.⁹ Bring your full medication and supplement list, including anything over-the-counter, to your next appointment and ask specifically about anticholinergic burden and interactions.

Hearing. This one surprises most people. The 2020 Lancet Commission on dementia prevention identified hearing loss in midlife as one of the most significant potentially modifiable risk factors across the life course.¹⁰ A 2023 analysis using nationally representative U.S. data quantified this directly: among adults with moderate or greater hearing loss measured by audiometry, an estimated 17 percent of dementia cases were attributable to that hearing loss, roughly eight times higher than estimates based on self-reported hearing problems alone.¹¹ A hearing screen is quick, painless, and one of the more overlooked items on this entire list, especially if you are over 50 or have worked around consistent loud noise.

Putting the list to work, at an educational level. None of these eight items requires specialized functional testing to check. Bring the worksheet near the end of this book to your primary care visit and ask, specifically, for each one to be addressed rather than assumed normal. Finding one genuinely treatable cause on this list is often worth more than months of supplements aimed at a problem that was never the real driver.

A Note on Popular Mood Supplements

Two over-the-counter supplements come up constantly in conversations about mood: St. John's Wort and 5-hydroxytryptophan (5-HTP). Both have real, studied effects on serotonin activity, which is exactly why both carry a serious interaction risk that deserves more attention than it usually gets. St. John's Wort has documented, clinically significant interactions with SSRIs and other antidepressants, oral contraceptives, blood thinners, and other prescription medications, including a risk of serotonin syndrome when combined with serotonergic antidepressants.¹² 5-HTP raises serotonin more directly, and combining it with an SSRI, SNRI, or MAOI carries a real risk of serotonin syndrome, a potentially serious reaction involving agitation,

rapid heart rate, tremor, and in severe cases, a medical emergency.¹³ Neither of these is a reason to panic if you have taken either one, but neither should be started, combined with a psychiatric prescription, or continued alongside one, without first talking to your prescriber. If you are not currently on a psychiatric medication and are considering either supplement, that conversation still belongs with a physician or my team before you start, not after.

The Gut-Brain Axis, and What Your Neurotransmitter Test Is Really Measuring

Where the science is real and the marketing runs ahead. The gut-brain axis is genuinely one of the more interesting and better-supported terrain concepts in this field, and it is also one of the most overstated in casual conversation and in some functional medicine marketing. A 2026 review describes the gut-brain axis as an integrative framework connecting gut microbial ecology, immune signaling, endocrine function, and mood regulation in depressive disorders, with mechanisms spanning short-chain fatty acid production, gut barrier integrity, and inflammatory signaling.¹⁴ That is real science. It is also a genuinely complex, still-developing field, not a settled explanation for any one person's symptoms.

What is actually happening in the body. Your gut lining, when healthy, is selective about what it lets into your bloodstream. When that barrier is compromised, a state sometimes loosely called “leaky gut,” bacterial byproducts and inflammatory signals can reach systemic circulation more easily, and systemic inflammation has real, documented effects on mood, which Key 5 covers in more depth. Your gut also produces the large majority of your body's serotonin, in the enterochromaffin cells lining your intestines, and hosts trillions of bacteria that interact with nutrient absorption, inflammation, and the vagus nerve, the primary communication line between your gut and your brain.

The caveat that comes with neurotransmitter testing. If you have had, or are considering, a urinary neurotransmitter test through a functional medicine practice, here is the straightforward picture. These panels measure real chemicals in a real sample, and the lab work itself is not the issue. The interpretation is where claims tend to run ahead of the evidence. Most of the serotonin your body makes is produced in your gut, and serotonin does not meaningfully cross the blood-brain barrier in either direction under normal conditions. The dopamine, norepinephrine, and epinephrine measured in a urine sample come overwhelmingly from your adrenal glands, your gut, and peripheral nerve activity, not from inside your skull. A detailed 2011 scientific review that specifically examined whether urinary neurotransmitter measurements reflect nervous system activity concluded that the strongest, most clinically established use for this kind of testing is detecting a rare adrenal tumor called pheochromocytoma, a use that has nothing to do with mood or cognition, and that the evidence connecting urinary neurotransmitter levels to depression, ADHD, or brain fog specifically was, in that review's own words, early and limited rather than settled.¹⁵ None of this makes the test worthless. A number on a urine panel is one data point, collected outside your brain, and it should never be handed to you as if it is a direct measurement of your brain chemistry, and it should never be the sole basis for a diagnosis or a treatment plan.

How to use this, at an educational level. If gut symptoms (bloating, irregular bowel habits, reflux, food reactions) travel alongside your brain fog or mood symptoms, that pattern is worth mentioning to your practitioner specifically, because gut-focused evaluation may be genuinely relevant terrain. If you are offered

a urinary neurotransmitter panel, ask what specific decision it will change, and hold its results alongside your actual symptoms and your other labs rather than as a stand-alone verdict on your brain chemistry.

Blood Sugar and the Aging Brain

The overlooked link between blood sugar and thinking. Blood sugar swings and insulin resistance are usually framed as a diabetes conversation, not a brain conversation, but your brain is exquisitely sensitive to both. Type 2 diabetes is increasingly recognized as a major risk factor for cognitive decline that extends well beyond its traditional metabolic complications.

The biology, without the jargon. A 2026 review describes cognitive decline associated with type 2 diabetes as arising from a convergence of gut dysbiosis, chronic low-grade inflammation, impaired insulin signaling in the brain itself, blood-brain barrier dysfunction, and mitochondrial abnormalities, a network the authors call the gut-brain-mitochondria axis.¹⁶ Insulin does more in the brain than regulate blood sugar. It participates directly in synaptic signaling and neuronal survival pathways, and when insulin signaling is chronically impaired, whether from prediabetes, insulin resistance, or diabetes itself, brain cells can become functionally starved for energy even when overall blood sugar looks acceptable on a single fasting test.

How sugar damage physically reaches the brain. There is one mechanism worth knowing by name, because it turns a vague association into something concrete. When blood sugar runs high, sugars bind to proteins and fats in a process called glycation, and the products of that reaction are called advanced glycation end products, or AGEs. AGEs act on a receptor called RAGE, the receptor for advanced glycation end products, and RAGE is the point where a metabolic problem becomes a neurological one. A 2020 review describes RAGE in Alzheimer's disease as both an intermediary of inflammation and an inducer of oxidative stress, involved in amyloid-beta production and clearance, in synaptic impairment, and in the two-way traffic between inflammation in the body and inflammation in the brain.¹⁷ The useful takeaway is not a product to buy. It is that the glycation load produced by years of high blood sugar has a well-studied route into brain tissue, which is why blood sugar keeps turning up in brain research that has nothing to do with diabetes care.

A term you are going to hear. Somewhere in this territory you will run into “type 3 diabetes” used as a nickname for Alzheimer's disease. It comes from a 2008 review that gathered the evidence for impaired insulin signaling in the Alzheimer's brain and argued that the pattern was distinct enough to deserve its own name.¹⁸ The observation underneath the nickname is real, and it is the same one described above. What the nickname is not is a diagnosis. There is no such condition on any diagnostic list, no physician will ever diagnose you with it, and the version that circulates online, that Alzheimer's simply is diabetes of the brain and that cutting carbohydrates therefore prevents it, claims considerably more than the research supports. Treat it as shorthand for a real line of investigation rather than as a settled answer, and be careful with anyone selling you a protocol on the strength of the nickname alone.

Where to start, at an educational level. Fasting glucose, hemoglobin A1c, and fasting insulin are worth adding to the labs from Key 2 if blood sugar has not been checked recently, particularly if brain fog is worse after meals or improves with regular eating. Beyond testing, the interventions with the most established evidence for stabilizing blood sugar (regular meal timing, adequate protein and fiber, and the movement

described in Key 8) are the same interventions that show up repeatedly throughout this book, which is exactly the point: a small number of terrain-level changes tend to move several of these dials at once.

Inflammation, the Quiet Driver of Mood Symptoms

The third contributor most conversations skip. Depression is still commonly discussed as a purely chemical or purely psychological problem. There is a substantial and growing body of research pointing to a third contributor: systemic inflammation.

How inflammation reaches mood, in plain language. A landmark 2016 review in Nature Reviews Immunology laid out the evidence that inflammatory cytokines, the signaling molecules your immune system uses to coordinate a response to infection or injury, can directly produce depressive symptoms when chronically elevated, through effects on neurotransmitter metabolism, neuroendocrine function, and neural circuits involved in reward and motivation.¹⁹ Two claims this evidence does not make: that all depression is inflammatory, and that anti-inflammatory supplements replace antidepressants or therapy. What it does say is that chronic, low-grade inflammation, whether from gut permeability, poor sleep, chronic stress, obesity, or an autoimmune process, is a real, biologically plausible contributor to mood symptoms in at least a meaningful subset of people, and it helps explain why addressing terrain-level inflammation sometimes improves mood symptoms that have not responded fully to standard treatment alone.

What you can do with this, at an educational level. A high-sensitivity CRP level is a simple, inexpensive screening marker for systemic inflammation. It is nonspecific: it tells you whether your body is carrying an inflammatory load, not where the load is coming from. That makes it a starting question rather than an answer, and it is worth discussing with your practitioner if fatigue, mood symptoms, and brain fog have traveled together for you. The nutrition and movement patterns in Key 8, along with the sleep and stress content in Keys 6 and 7, are the best-supported levers for lowering chronic inflammation without medication.

Sleep, Your Brain's Nightly Cleaning Shift

A cause disguised as a symptom. Poor sleep is treated as a symptom of brain fog and low mood. It is at least as accurate to treat it as a cause, because of a specific, physical cleaning process that depends on sleep to run properly.

The nightly cleaning process, explained simply. A landmark 2013 study in mice, published in the journal *Science*, discovered a system now called the glymphatic system: a network that clears metabolic waste, including proteins linked to neurodegeneration, out of brain tissue. The researchers found that natural sleep was associated with a striking increase in the space between brain cells, allowing fluid to flow through and flush out waste products far more efficiently than during wakefulness.²⁰ That finding was in mice, and the exact scale of the effect in humans is still an active area of research, but the core concept, that sleep is doing active maintenance work in your brain and not simply “resting,” is now well established and helps explain why poor sleep so reliably produces next-day brain fog, regardless of what else is going on in your health.

Practical steps, at an educational level. Consistent sleep and wake times, a dark and cool bedroom, and limiting alcohol close to bedtime (alcohol fragments sleep architecture even when it helps you fall asleep faster) are the highest-yield, lowest-cost changes here. If you snore heavily, wake gasping, or feel unrefreshed no matter how many hours you log, revisit the sleep apnea section in Key 2 rather than assuming the fix is simply more time in bed.

Chronic Stress and the HPA Axis

Why “stress” gets dismissed, and should not be. “Stress” is such a broad, overused word that it has become easy to dismiss as an explanation for brain fog and mood symptoms. The biology underneath chronic stress is specific, measurable, and directly relevant to both.

What chronic stress does physically. Your hypothalamic-pituitary-adrenal (HPA) axis is the signaling chain that governs your cortisol response to stress. A 2025 review describes how prolonged activation of the HPA axis, from chronic (not occasional) stress, disrupts normal cortisol regulation and interacts with inflammatory pathways to produce oxidative stress and structural and functional changes in the hippocampus, the brain region most central to memory formation and emotional regulation.²¹ Nobody here is saying stress is “all in your head,” or that willpower can simply override it. The research describes a real, physical feedback loop: chronic stress changes hippocampal function, and a dysregulated hippocampus makes it harder to regulate the stress response, which can compound both memory difficulty and mood symptoms over time if the loop is not interrupted.

Breaking the loop, at an educational level. Interrupting a chronic HPA axis pattern is rarely about eliminating stress, which usually is not realistic, and much more about building in genuine nervous-system downtime: consistent sleep (Key 6), regular movement (Key 8), and practices such as slow breathing, time outdoors, or meditation that have research support for lowering cortisol reactivity. If stress feels unmanageable or is tied to a specific trauma, a therapist is not a lesser option than a supplement protocol here. It is often the more direct and better-supported one, and pairs well with everything else in this book rather than competing with it.

Nutrition and Movement, the Two Levers With the Strongest Evidence

The least exciting advice, with the strongest evidence. Of everything in this book, the interventions with the most rigorous supporting evidence are also the least exciting to hear about: what you eat on a normal Tuesday, and whether you move your body regularly. Neither requires a lab test or a supplement, and both are frequently the first thing to slip when brain fog and low mood make everything feel harder.

What the trials actually found. A large 2015 cohort study tracking older adults over several years found that people whose diets most closely followed the MIND diet, a hybrid of the Mediterranean and DASH dietary patterns emphasizing vegetables (especially leafy greens), berries, nuts, whole grains, fish, and olive oil, showed a substantially slower rate of cognitive decline than those whose diets least resembled it, a difference in decline rate the researchers described as equivalent to being 7.5 years younger.²² That was an observational study, not a controlled trial, so it shows a strong association rather than definitive proof, and the researchers themselves noted that a dietary intervention trial would be needed to confirm the finding, but the size and consistency of the association make it one of the better-supported nutrition patterns in this entire field. On the mood side, a randomized controlled trial called SMILES tested whether a structured dietary improvement program, delivered by a clinical dietitian alongside participants' existing therapy and medication, could improve depression symptoms. It found a significantly greater improvement in depression scores in the dietary support group compared with a social support control group, with roughly one in three participants in the dietary group reaching remission compared to one in twelve in the control group.²³ This was tested as an addition to standard care, not a replacement for it, which is exactly how to think about nutrition's role in this book.

On movement, a randomized controlled trial of 120 older adults found that a year of aerobic exercise training increased the volume of the hippocampus by about 2 percent, offsetting the equivalent of one to two years of the age-related shrinkage that region normally undergoes, alongside measurable improvements in memory.²⁴ And in the most rigorous trial to date testing a combined lifestyle approach, the FINGER trial randomly assigned older adults already at elevated dementia risk to either a two-year multidomain program (diet, exercise, cognitive training, and vascular risk monitoring) or general health advice. The intervention group showed a modest but statistically significant improvement in a composite cognitive test score compared with the control group.²⁵ The effect was real but modest, this was a "proof-of-concept" trial in older adults already flagged as at-risk, and it does not mean cognitive decline or dementia can be reliably prevented or undone in every person. Still, a combined approach, tested rigorously, measurably moved a hard outcome in a real trial, which is a meaningfully stronger claim than any single supplement or diet trend can currently make.

The order that works, at an educational level. If you take one thing from this entire book, let it be this: a genuinely Mediterranean-leaning or MIND-style eating pattern and regular aerobic movement, three or more

times weekly, are the two changes with the deepest and most consistent evidence behind them for both brain fog and mood, and they are also the two most commonly skipped in favor of a more targeted-sounding supplement. Layering supplements on top of a poor foundation is a common and expensive mistake. Building the foundation first, then adding anything else on top of it, is the order that the evidence actually supports.

Chronic Infection, Toxicant Exposure, and the Longer-Horizon Picture

The longer-horizon question readers eventually ask. Everything so far in this book addresses brain fog and mood you are noticing now. This chapter turns to a different, longer-horizon question that a lot of readers eventually ask anyway, often after watching a parent's memory decline or after reading something online: does anything I am exposed to now affect my risk many years from now? There is a real, mainstream, still-developing body of research on modifiable contributors to long-term cognitive risk, and it deserves an honest answer rather than either dismissal or overclaiming. One boundary bears restating before going further: this book remains written for brain fog and mood, not for diagnosed dementia, Alzheimer's disease, or Parkinson's disease, and no item below carries a claim that addressing it will prevent or reverse a neurodegenerative disease. What follows is a longer list of terrain worth understanding and, where reasonable, addressing, in the same spirit as the gut function, blood sugar, sleep, and stress content already covered.

Chronic infection. The idea that infections contribute to Alzheimer's risk has moved from a fringe theory to genuinely active, mainstream research in the last few years. A 2025 systematic review and meta-analysis pooling 26 studies and more than 1.2 million participants found herpes simplex virus (HSV) infection associated with a meaningfully higher risk of Alzheimer's disease: about 32 percent higher in case-control studies and 20 percent higher in cohort studies, with HSV-1 specifically associated with 46 percent higher odds.²⁶ A separate 2026 study following more than 8,500 participants in the United Kingdom for up to 24 years tested three infections directly, *H. pylori*, HSV-1, and varicella-zoster virus (the virus behind chickenpox and shingles), and found a significant association with dementia risk for *H. pylori* specifically (24 percent higher risk), while HSV-1 and varicella-zoster did not reach statistical significance in this particular cohort.²⁷ Different studies finding different infections significant is normal in an active research area, not a sign the whole field is unreliable, which is exactly why this book cites more than one study rather than leaning on a single result. On the treatment side, a large 2026 retrospective study of more than 230,000 matched patients found antiviral treatment for oral or facial HSV infection associated with about 13 percent lower risk of Alzheimer's disease and 17 percent lower risk of dementia, though the effect was not seen for genital HSV, and the authors themselves call for prospective confirmation rather than treating the finding as settled.²⁸

Lyme disease specifically, the tick-borne infection Dr. Bredesen and others have pointed to as a contributor worth investigating, has a thinner and more mixed evidence base than the herpesvirus research above. A small 2019 case-control study in Mexico compared 108 cognitively healthy older adults with 38 people with Alzheimer's and 39 with mild cognitive impairment, and found *Borrelia burgdorferi* (the bacterium that causes Lyme disease) antibodies in 29 percent of the Alzheimer's group compared with 10 percent of healthy controls, a significant association after adjustment.²⁹ A larger, more recent 2026 systematic review pooling studies on spirochetal infections, the bacterial family that includes Lyme disease, syphilis, and leptospirosis,

found the picture for Lyme disease specifically to be heterogeneous and inconclusive, in contrast to more consistent findings for syphilis and leptospirosis.³⁰ Where the evidence nets out: chronic infection as a category, and herpesviruses and *H. pylori* specifically, has real and mounting support behind it. Lyme disease's specific role is a reasonable question worth further research, but is not yet established the way the herpesvirus findings are. If ongoing or past infection, including Lyme disease, Epstein-Barr virus, or a lingering post-viral pattern, feels like it fits your own history, our companion guide *Unlock Your Post-Viral Recovery* goes considerably deeper into immune activation and chronic infection than this book does.

Toxicant and environmental burden. This is a category where honesty about exposure level matters more than almost anywhere else in this book. Heavy metal neurotoxicity is real, but the strongest evidence comes from occupational or unusually high-dose exposure, not typical background exposure. A 2020 review of manganese, an essential nutrient at normal dietary levels, found that industrial and occupational overexposure (welding and mining are the classic examples) is linked to oxidative stress, neuroinflammation, and disrupted brain transporter function, a very different exposure picture than getting manganese from food.³¹ Mercury shows a similar pattern. A 2021 review of Amazon river communities living near artisanal gold mining found measurable cognitive, motor, and emotional deficits in populations with mercury body burden two to ten times recommended limits, while a separate review focused on dental personnel, people occupationally exposed to mercury vapor while handling amalgam, found associations with markers of neurotoxicity.³²³³ Both describe meaningfully elevated, ongoing occupational or environmental exposure, not the low-level exposure most readers have from a filling or an occasional meal of fish, and this book will not imply otherwise. If you have a specific occupational exposure history (mining, welding, dental work, or other industrial work) or live near a known contamination source, that history is worth mentioning to your practitioner directly.

Mold and mycotoxin exposure is a related concern that comes up often in functional medicine conversations, including Dr. Bredesen's. A 2018 review describes reported effects of mold and mycotoxin exposure across multiple body systems, most consistently respiratory and allergic symptoms, with nervous-system effects described as a real but considerably less developed area of research.³⁴ A specific search for evidence connecting mold exposure to long-term dementia risk did not turn up a human evidence base comparable to what exists for the infections above, and this book will not overstate a connection the research does not yet support.

Air pollution has the strongest and most consistent evidence of any environmental exposure in this chapter. A 2025 systematic review and meta-analysis pooling data across tens of millions of participants found long-term exposure to fine particulate matter (PM2.5) associated with about 8 percent higher dementia risk per 5 micrograms per cubic meter of average exposure, with positive associations also found for nitrogen dioxide and black carbon exposure, while some other pollutants (nitrogen oxides, PM10, ozone) showed no clear association.³⁵ These are modest, population-level effect sizes, not a claim that your personal air quality determines your cognitive future, but they are consistent enough across a large evidence base that air pollution is one of the named factors in the Lancet Commission framework below. If reducing ongoing toxicant exposure interests you, our companion guide *Unlock Your Detox Capacity* covers this in depth,

including the honest distinction between validated, unprovoked testing and the provoked or chelation-based approaches I do not use.

Head injury. A history of significant head injury, even a single moderate or severe traumatic brain injury, is a recognized contributor to later dementia risk, and it deserves brief, honest mention even though little can be done retroactively about an injury that already happened. A 2023 meta-analysis combining civilian and military data found a history of traumatic brain injury associated with about 66 percent higher risk of later dementia, an association that did not differ meaningfully by injury severity or between veterans and civilians, though veterans as a group carry a larger population-level impact simply because head injury is more common in that population.³⁶ If you have a history of concussion, a more serious head injury, or repeated head impacts from contact sports or another cause, that history is worth mentioning to your practitioner as part of your overall picture, not as a cause for alarm on its own.

The mainstream umbrella: the Lancet Commission. Everything above can read like a scattered list of separate worries. The most authoritative attempt to organize modifiable dementia risk factors into one evidence-based framework is the Lancet Commission on dementia prevention, intervention, and care, an international panel of dementia researchers whose most recent update was published in August 2024 in *The Lancet*, following the widely cited 2020 report already referenced in Key 2 of this book.³⁷ Its named modifiable risk factors include hearing loss (Key 2 above), physical inactivity and diet (Key 8 above), and the air pollution and traumatic brain injury just discussed, among others spanning early life, midlife, and later life. Two things are worth saying plainly. First, this is genuinely mainstream, non-fringe work, published in one of medicine's most rigorous journals and reviewed by dozens of dementia specialists worldwide, not an alternative or functional medicine framework. Second, the Commission's own position is that a substantial share, not all, of dementia risk is potentially modifiable, and addressing every factor on its list does not guarantee any individual outcome. This book leans on the Commission because it is the legitimate, thoroughly vetted version of the same instinct that draws people toward more speculative frameworks: the idea that cognitive risk over a lifetime is not fixed at birth and has real, addressable contributors.

A Note on Dr. Bredesen's Multifactorial Approach

If you have read about Dr. Dale Bredesen's work, this is a fair place to address it directly, since the terrain covered in this chapter overlaps with territory he has written about for years. Dr. Bredesen is a neurologist and researcher affiliated with UCLA's David Geffen School of Medicine who has argued that cognitive decline usually has multiple simultaneous contributors, including chronic infection and toxicant exposure among others, rather than one single cause, and that a personalized workup addressing several contributors at once may help more than treating cognitive decline as a single disease process. That underlying instinct, that cognitive decline is often multifactorial and that infection and toxicant exposure belong on the list of things worth investigating, is consistent with the mainstream research already described above. Where this book draws a firm line is around his specific published clinical evidence to date. The primary published trial of his personalized, multi-domain protocol enrolled 25 patients with dementia or mild cognitive impairment in a single-arm design with no control group, described by its own authors as a proof-of-concept study, and its own authors explicitly call for a larger randomized controlled trial before the approach can be considered

established.³⁸ That is an honest, preliminary starting point for further research, not the same thing as a proven protocol. This book does not adopt his specific diagnostic framework or terminology. What it does share with his work is the well-supported underlying idea that chronic infection and toxicant exposure belong on the list of things worth understanding about long-term brain health, which is exactly why they appear above.

A Note on Statins and Brain Health

Some clinicians, Dr. Bredesen among them, advise avoiding statin medications specifically for brain health reasons. That advice deserves a careful look, not dismissal and not alarm. In February 2012, the FDA did add a label change to statin medications noting rare post-marketing reports of cognitive effects, memory loss, forgetfulness, and confusion, generally reversible within weeks of stopping the medication.³⁹ That label change is real and worth knowing about. But it does not mean statins are, on balance, harmful to the brain. A large 2022 meta-analysis pooling 36 observational studies found statin use associated with about 20 percent lower odds of dementia and 32 percent lower odds of Alzheimer's disease specifically, with no meaningful difference between lipophilic statins (which cross the blood-brain barrier more readily) and hydrophilic statins (which do not), a finding that directly undercuts the theory that statins harm cognition by crossing into the brain.⁴⁰ A separate 2021 study following nearly 19,000 older adults for almost five years found no association between statin use and incident dementia, mild cognitive impairment, or cognitive decline.⁴¹ Memory complaints are a real, labeled, reversible effect reported by a small number of people taking statins, but they are not a common side effect, and the weight of current evidence does not show statins increasing dementia risk. As with every medication decision in this book, the answer is never to stop or reduce a statin on your own. If you are concerned about a statin's effect on your memory or thinking, that is a conversation to have with the physician who prescribed it, exactly the way this book approaches every other prescription decision.

How to hold all of this, at an educational level. None of this changes the boundary stated at the start of this book. It remains written for brain fog and mood, not for diagnosed dementia, Alzheimer's disease, or Parkinson's disease. Addressing any of these contributors comes with no promise of preventing or reversing a neurodegenerative disease in you or anyone else. What this chapter offers is an honest, mainstream-grounded answer to a question many readers eventually ask: if you are thinking about your brain health over decades, not just this week, what is actually worth knowing about? Reducing ongoing toxicant exposure where that is reasonable, treating a genuine chronic infection if one is found, protecting your head during activities with real injury risk, and knowing your own occupational or environmental exposure history are all reasonable, low-risk things to know and to mention to your practitioner. None of them replace the higher-certainty, more immediately actionable items in Key 2, and none of them are a reason to start an expensive testing protocol without a specific reason to suspect it applies to you.

A Staged Roadmap

The first two weeks. Bookmark the crisis resources at the front of this book. Start the symptom and sleep tracker below so you have a real baseline, not a memory of one. Schedule an appointment with your primary care provider or my team and bring the treatable-causes checklist below. If you are on a psychiatric medication, change nothing about it on your own.

Weeks three through eight. Begin shifting your eating pattern toward the MIND-style approach in Key 8, and add regular aerobic movement if it is not already part of your routine, starting wherever your current fitness allows. Address whatever came back on your labs from Key 2: correct a real deficiency, treat a real thyroid abnormality, start sleep apnea evaluation if indicated, and review your medication list for anticholinergic burden. Tighten sleep and wake times per Key 6.

Ongoing. Continue tracking. If brain fog or mood symptoms are not meaningfully better after 8 to 12 weeks of addressing the treatable causes and building the nutrition and movement foundation, that is useful information, not a failure, and it is the point at which more specialized testing (gut-focused, inflammatory, or otherwise) becomes genuinely worth considering with a practitioner, rather than being the first move.

Your Symptom and Sleep Tracker

Rate each item once daily for at least two weeks before making any change, so you have a real baseline to compare against. Repeat the same tracker after 30 and 90 days of any new approach to see whether it actually helped, since memory alone is a notoriously unreliable way to judge gradual change.

Date	Hours slept	Sleep quality (1 to 5)	Brain fog (1 to 10)	Mood (1 to 10)	Stress (1 to 10)	Caffeine after 2pm?	Alcohol?	Movement today?	Notes

Your Treatable-Causes Checklist

Bring this list to your next medical appointment and ask that each item be specifically addressed rather than assumed normal.

- ☐ Vitamin B12, and ideally methylmalonic acid (MMA), a more sensitive marker
- ☐ Folate, ideally red blood cell folate rather than serum alone
- ☐ TSH and free T4 (and free T3 if your practitioner agrees it is warranted)
- ☐ Complete blood count (CBC) and ferritin
- ☐ Vitamin D (25-OH)
- ☐ A full medication and supplement list, reviewed specifically for anticholinergic burden and interactions
- ☐ A conversation about snoring, witnessed breathing pauses, or daytime sleepiness, with a referral for a sleep study if indicated
- ☐ A hearing screen, especially if you are over 50 or have significant noise exposure history
- ☐ Fasting glucose, hemoglobin A1c, or fasting insulin, if blood sugar has not been checked recently
- ☐ High-sensitivity CRP, if inflammation has not been assessed and mood, fatigue, and brain fog are traveling together

Take the Next Step: Your Free Root Discovery Session

The Root Discovery Session is a complimentary consultation with my team to review your specific symptoms, which of the treatable causes in this book have and have not been checked, and whether a more personalized workup makes sense for your situation. There is no obligation, and everything in this conversation is meant to run alongside, not replace, any mental health or medical care you are already receiving.

Schedule your free Root Discovery Session at rootcauseunlocked.com/discovery.

You can also reach the office directly at (919) 662-0520.

Not ready for a conversation yet? Start tracking.

The tracker earlier in this guide works on paper. The digital version does the same job on your phone, keeps your history in one place, and shows you the connections that are hard to see from inside them: how sleep tracks against next-day clarity, and whether mood and focus move together or separately. Two weeks of honest data makes any future appointment, with our team, your physician, or your mental health provider, dramatically more useful than memory alone.

Start your free tracker at rootcauseunlocked.com/tracker.

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If you are having thoughts of suicide or self-harm, please call or text 988, call 911, or go to your nearest emergency department before doing anything else.

This guide is provided for general educational purposes and does not constitute individualized medical, psychiatric, or psychological advice, diagnosis, or treatment, and it does not diagnose dementia, Alzheimer's disease, depression, anxiety, or any other condition. It is not a substitute for a relationship with your physician, psychiatrist, therapist, or other qualified health care provider. If you are having thoughts of suicide or self-harm, call or text 988, call 911, or go to your nearest emergency department immediately. Never stop, start, reduce, or otherwise change the dose of any prescription medication, including antidepressants, anti-anxiety medications, or mood stabilizers, without first talking to the prescriber managing your care. Always consult your physician before starting any new supplement, particularly if you are pregnant, nursing, taking prescription medications, or being treated for a psychiatric condition.

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